QUANT RESEARCH / STATISTICS

ALPHA VALIDITY

07

PROPOSED RESEARCH / NOT YET VERIFIED

THE ENGINEERING QUESTION

How well do common validation methods distinguish a real predictive signal from a chance pattern?

PROBLEM DEFINITION AND RESEARCH CONTRIBUTION

A scientific investigation into false discoveries in strategy research, measuring how selection bias, overfitting and costs distort apparent alpha.

- The point is methodological honesty, not presenting a flashy equity curve.
- Strong negative results can be as impressive as a profitable toy backtest if the methodology is rigorous.
- This should prove statistical judgement more than finance theatre.

ARCHITECTURE AND IMPLEMENTATION SPECIFICATION

1 STAGE **Synthetic markets** → 2 STAGE **Signal search** → 3 STAGE **Validation lab** → 4 STAGE **Statistical report**

- Return/process generators with null and weak signal dynamics, plus explicit regime shift scenarios.
- Research engine for signal search, walk forward testing, bootstrap intervals and multiple testing controls.
- Reporting pipeline that emphasises calibration, selection bias and economically realistic conclusions.

ENGINEERING DECISIONS TO MAKE EXPLICIT

The design review must record data contracts, state transitions, failure handling and why the selected baseline is defensible. Every subsystem should have an interface specification and a reproducible test route.

DESIGN BOUNDARIES AND ASSUMPTIONS

⚠ Synthetic market results do not prove tradable real world alpha.

⚠ Survivorship bias, leakage and market data licensing must be explicitly controlled.

ⓘ Scope is proposed unless explicitly identified as documented existing work. All measured claims require reproducible evidence.

ALPHA VALIDITY / PROTOCOL / ACCEPTANCE / REVIEW

EXPERIMENTS, VERIFICATION & PUBLICATION

07

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EXPERIMENTAL METHODOLOGY

01 Generate null and weak signal markets, validating the simulated properties first.

02 Simulate researcher selection from large strategy universes under multiple validation methods.

03 Measure false discovery, power, regime sensitivity and cost adjusted robustness on synthetic and licensed data.

MEASUREMENT AND ACCEPTANCE EVIDENCE

- Empirical false discovery rate and statistical power
- Sensitivity to regime shift and data snooping
- Synthetic market conclusions versus licensed historical data

REQUIRED REPOSITORY ARTEFACTS

- Reproducible statistical experiments
- Calibration plots and negative result reporting
- Technical paper and research notebook bundle

TECHNICAL REVIEW QUESTIONS

Which assumptions could invalidate the result? How do the baselines behave under the same conditions? What is the strongest negative case? Which variables, seeds, workloads, software versions and machine properties must be retained for independent reproduction?

VERIFICATION MAPPING

For each acceptance claim, record the test identifier, input configuration, observed result, pass/fail definition and an independent rerun procedure. Failed or inconclusive cases remain in the report rather than being silently excluded.

PUBLICATION DISCIPLINE

Release original source, configuration files, test harnesses, raw measurements and scripts where rights permit. Distinguish simulation outcomes from physical validation and completed achievements from planned evaluation. Results must not be inferred from targets.